

MOBILE APPLICATION DEVELOPMENT

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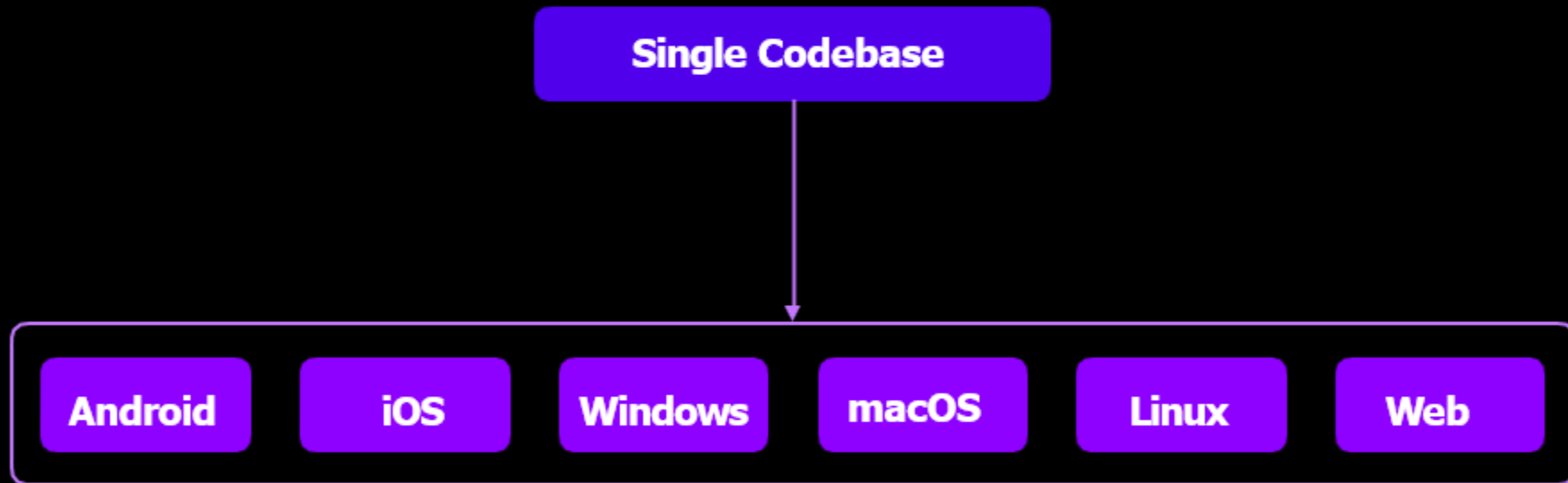
Instructor Information

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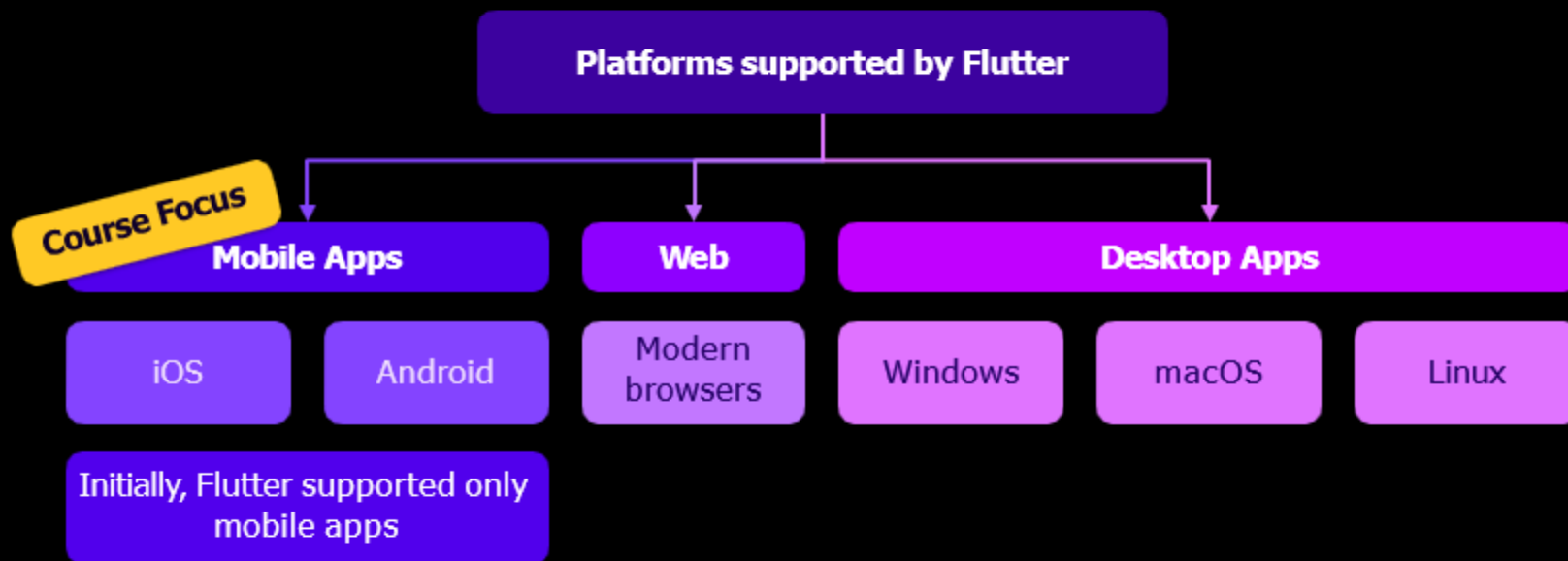
Let's Start

Lecture #2

One Codebase, Multiple Apps



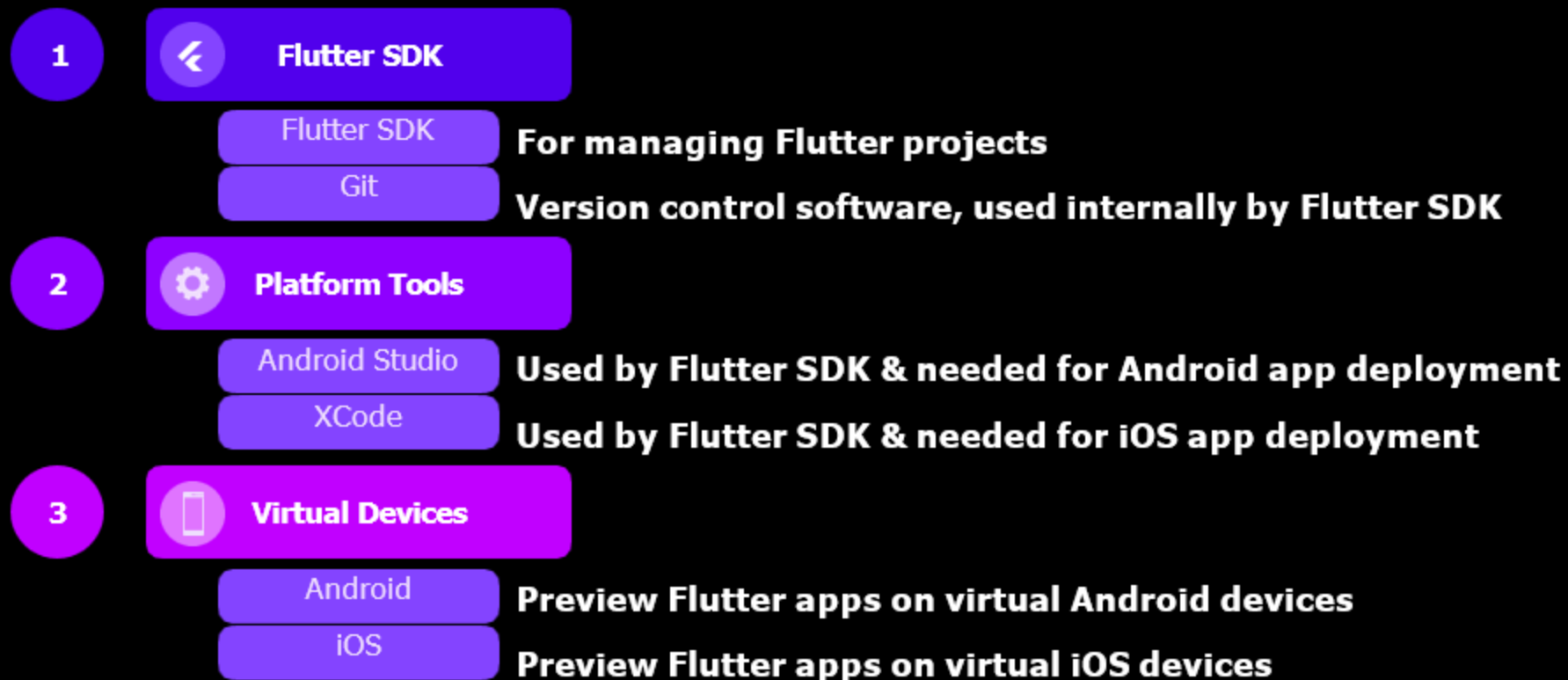
Target Platforms



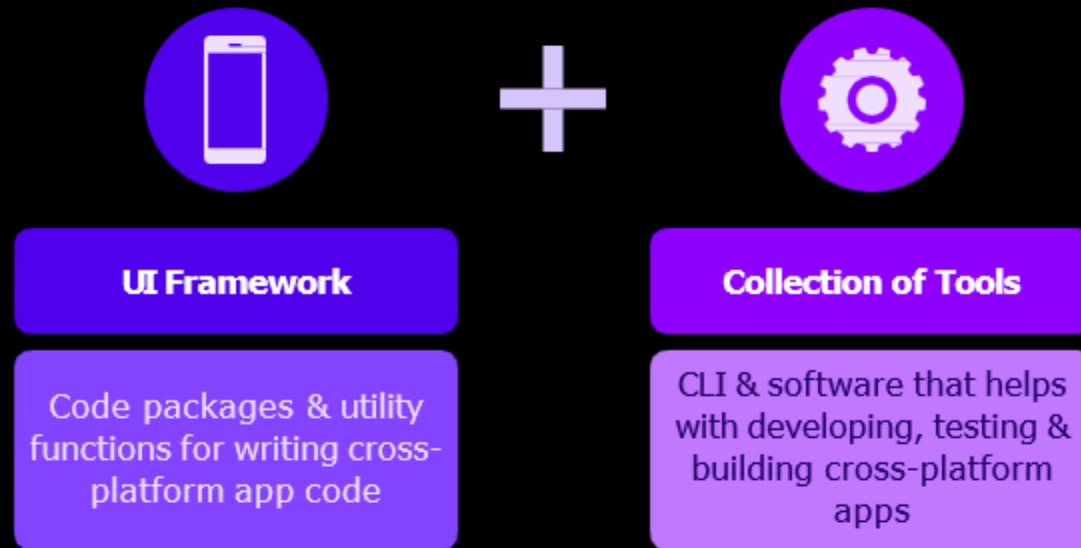
Whilst you can write the code for all platforms on the same machine, you can **only test & run iOS & macOS apps on macOS machines, Windows apps on Windows machines and Linux apps on Linux machines!**

Android and web apps can be built on all operating systems.

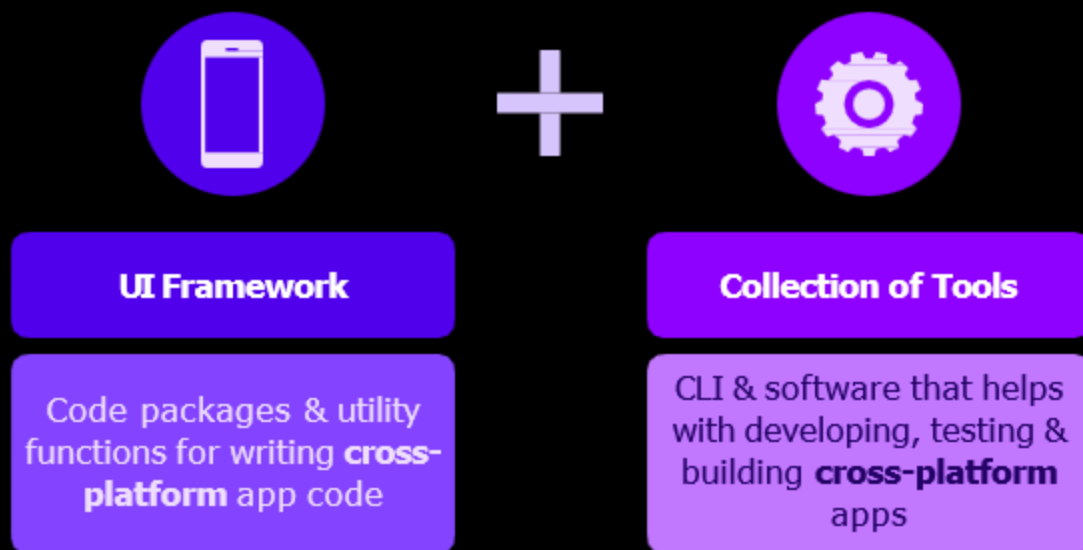
Flutter Setup



What Is Flutter?

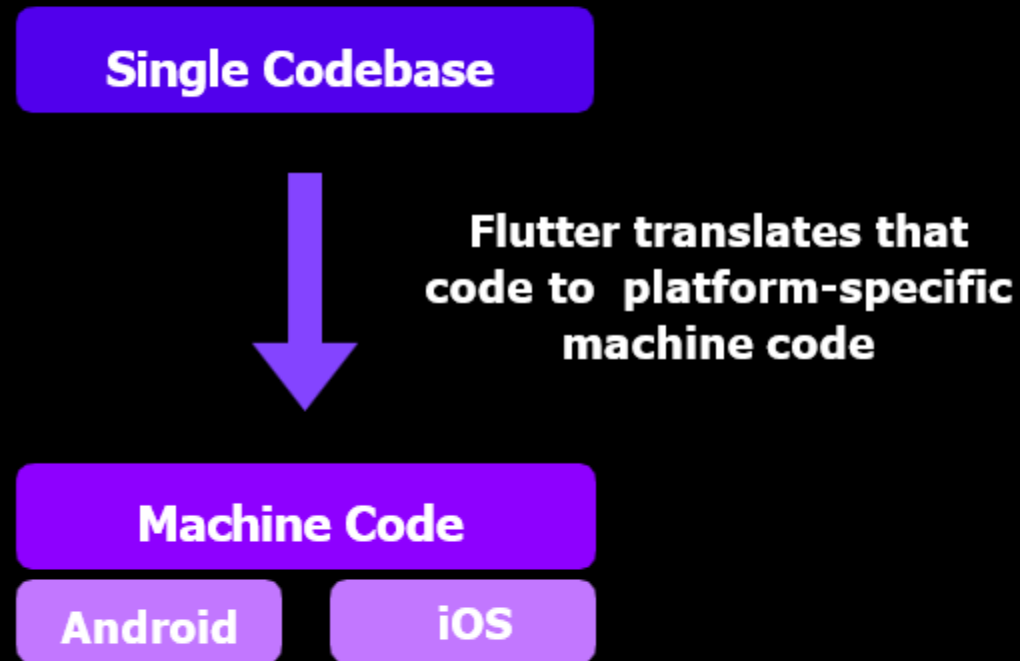


What Is Flutter?



Flutter allows you to build multi-platform apps based on one single codebase and programming language

From Flutter Code To Platform Code



Flutter Is Not A Programming Language!

It's a **framework** for building user interfaces with **Dart**

Framework

A collection of
packages &
utility
functions you
may use in
your code

Dart



A programming
language
developed by
Google

Main usage:
Flutter app
development

INTRODUCTION

Flutter, Google's open-source UI framework, has revolutionized how developers create cross-platform applications. Launched in 2018, Flutter allows developers to write a single codebase and deploy it across multiple platforms, including Android, iOS, web, and desktop. But is it the right choice for new developers?

Why Flutter is best for Mobile APP Development?



Flutter

- ✓ Flexible UI
- ✓ Modern, reactive Framework
- ✓ Native Performance
- ✓ Fast Development
- ✓ Highly Customisable

WHAT IS FLUTTER

Flutter is a framework powered by the Dart programming language. It enables developers to build highly responsive and visually appealing applications. With its “**write once, run anywhere**” approach, Flutter has become one of the most sought-after tools for startups and large enterprises alike.

ADVANTAGES OF FLUTTER

1. Cross-Platform Development

Flutter allows multi-platform app development from a single codebase, saving time, reducing costs, and ensuring consistent performance.

2. Hot Reload

Hot Reload shows changes instantly without restarting the app, speeding up development, debugging, and UI testing.

3. Rich Pre-Built Widgets

Flutter provides a wide range of customizable widgets, including Material and Cupertino, for creating adaptive, native-like Android and iOS apps.

ADVANTAGES OF FLUTTER

4. High Performance

Flutter compiles directly to native code, delivering faster startup and smoother performance than JavaScript-based frameworks.

5. Open Source and Strong Community Support

Flutter is open source with a strong community, offering many plugins, libraries, and learning resources.

6. Fewer Platform-Specific Issues

Flutter reduces platform-specific code using built-in APIs and plugins, making features like camera, GPS, and animations easy to integrate.

Why Flutter is So Popular?

- Swift Development
- Google Backing
- Singular Codebase
- Easy on Pocket
- Packages
- Additional animation & Widgets
- High Performance
- Secure
- Favor Social Networking



DISADVANTAGES OF FLUTTER

1. Large App Size

Flutter apps tend to have larger file sizes compared to native apps. This might be a drawback in markets where low-bandwidth devices are common.

2. Dart Language Adoption

Dart is not as popular as JavaScript, Java, or Kotlin. New developers might face a learning curve when switching to Flutter.

3. Performance on Web and Desktop

Flutter's performance on mobile is impressive, but its web and desktop capabilities are still maturing. Developers aiming for these platforms might encounter limitations.

SCOPE OF FLUTTER FOR NEW DEVELOPERS

- **Opportunities across e-commerce, healthcare, fintech, and education**
- **Strong freelancing and remote work potential**
- **Widely used in startups for cost-effective app development**
- **Continuous growth and support from Google**

Reasons to Choose Flutter Over React Native



	Native Apps	Hybrid Apps	Cross-Platform Apps
Development	Built for a specific platform (iOS or Android) using platform-specific languages (e.g., Swift/Objective-C for iOS, Kotlin/Java for Android).	Built using web technologies (HTML, CSS, JavaScript) and wrapped in a native container (WebView).	Built with a single codebase using frameworks (e.g., Flutter, React Native) that compile to near-native code.
Performance	Superior performance, speed, and responsiveness as they are optimized for a specific OS and have direct hardware access.	Generally slower performance due to the reliance on a WebView and an extra layer between the code and the OS.	Good, near-native performance, better than hybrid but generally slightly lower than fully native, especially for complex animations/graphics.

Thank You 😊